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MOOCS Decision Modeling Course

Simulation

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Module 21

Introduction to Simulation

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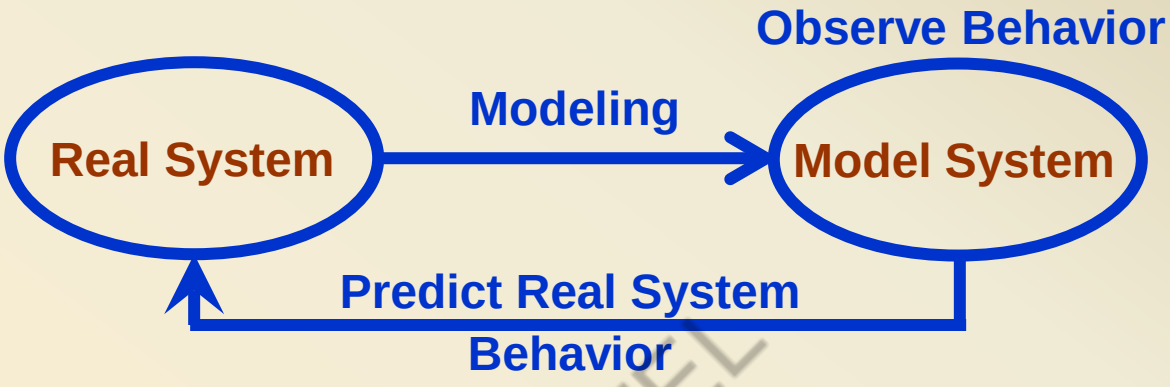
What is Simulation?



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Simulation includes developing a model of a real system and experimental manipulation of the model to predict real system behavior.

The real systems are often complex, dynamic and interactive.

A model is an abstraction of a real system representing the objects within the system and their interactions.

Why Use Simulation?

- Ability to experiment with model system rather than the real system
- Ability to obtain results in less time
- Ability to control sources of variations
- Ability to avoid errors in measurement
- Ability to generate alternate scenarios for decision making
- Ability to obtain replications

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Simulation Examples

Some examples of simulation of large real life systems:

- Simulation of the operations of a large airport to obtain the impact of policies on adding or deleting aircrafts, maintenance facilities, people, berthing facilities etc.
- Simulation of large scale distribution and inventory control systems to improve their design.
- Simulation of large scale military battles to evaluate defensive and attacking military systems.
- Simulation of a telephone communication system to find out component capacities to provide satisfactory service level at minimum possible costs.

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Types of Simulation Models

Discrete-Event Simulation Model

- ***Stochastic***: **random state variables exist**
- ***Dynamic***: **State changes with time**
- ***Discrete-Event***: **State changes at discrete time instances**

Monte Carlo Simulation Model

- ***Stochastic:*** **random state variables exist**
- ***Static:*** **State is not changing with time**

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Continuous Simulation Model

- ***Deterministic or Stochastic***: **Deterministic or random state variables**
- ***Dynamic***: **State changes with time**
- ***Continuous:*** **State changes continuously – guided by a set of differential equations**

Discrete Event and Continuous Simulation

Discrete Event Simulation	Continuous Simulation
Events occur	Events do not occur
Made up of *entities*, *attributes*, and *events*	Changes expressed in terms of differential equations
System state evolves at discrete points in time	System state evolves continuously in time
Random Numbers are required	Random Numbers are not required
Examples: Queuing system, inventory models, Machine Shop models	Examples: Econometric models, System Dynamics, Classical Mechanics

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Discrete Event Simulation

Discrete Event Simulation is made up of *entities, attributes, events, activities, and state variables.*

Entity: **Representation of real system object important in the context of the model**

Attribute: **Characteristic of an entity**

Event: **An interaction between entities**

Activity: **Actions related to entities and events**

State Variable: **Variables values representing system state**

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Discrete Event Simulation

Production System

- Entities: Machines
- Attributes: Speed, Capacity, Breakdown Rate
- Activities: Welding, Cutting, Turning etc.
- Events: Breakdown
- State Variables: Status of Machines- Busy, Idle, or Down

Inventory System

- Entities: Warehouse
- Attributes: Capacity
- Activities: Withdrawing, Purchasing
- Events: Demand
- State Variables: Inventory Levels, Backlogged Demands

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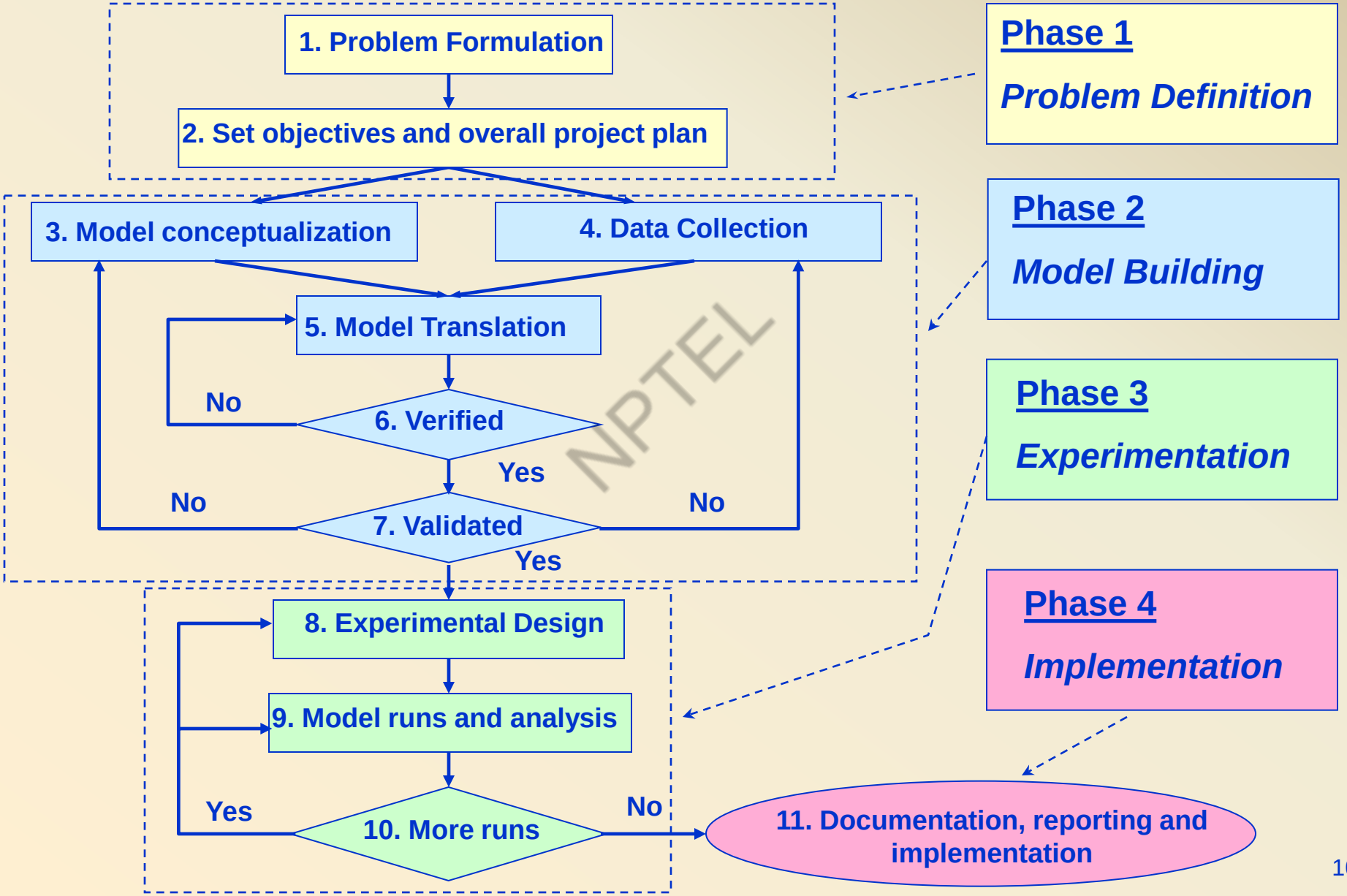
Discrete Event Simulation Steps



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Discrete Event and Monte-Carlo Simulation

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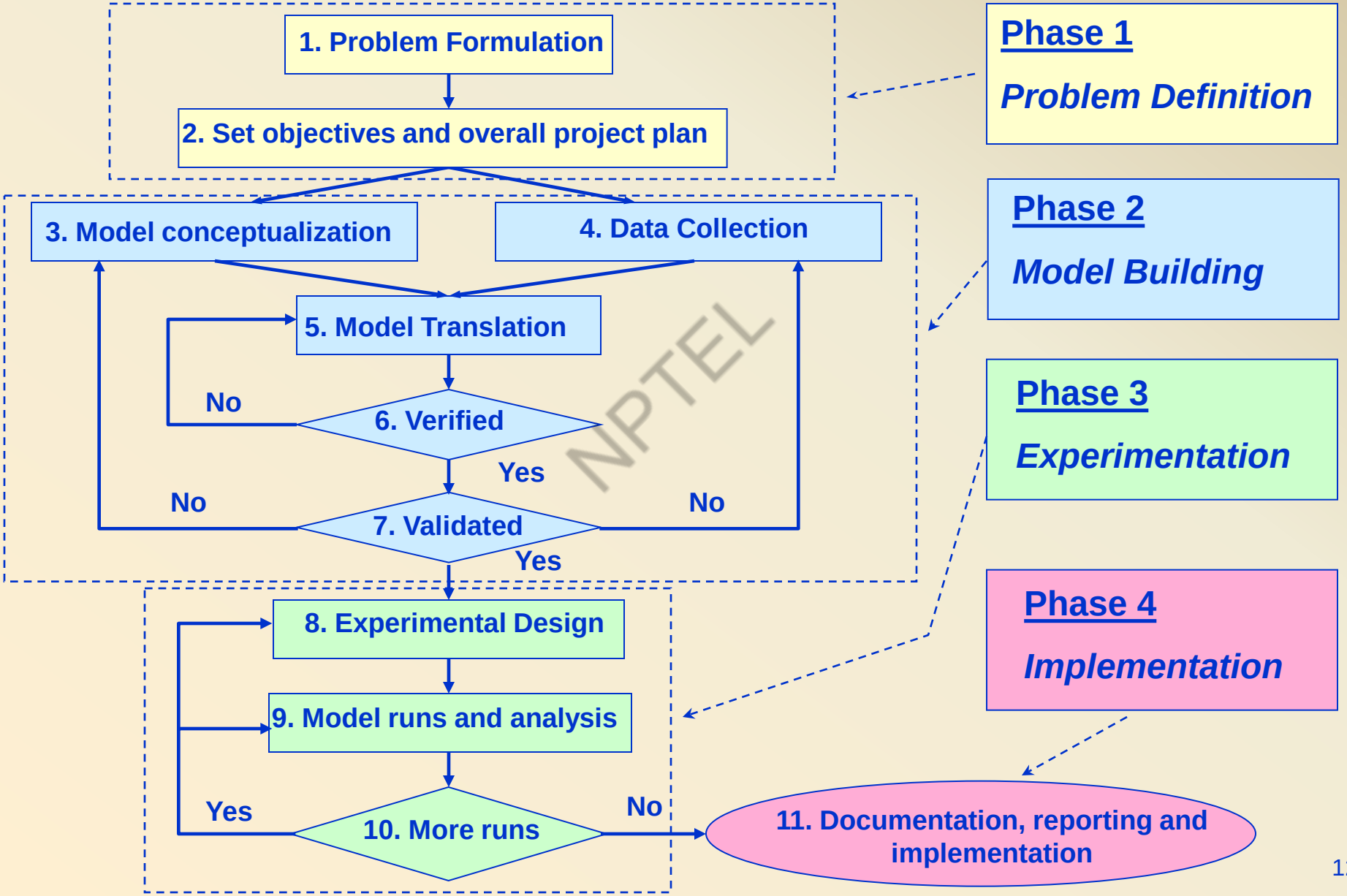
Discrete Event Simulation Steps



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Model Synchronization

A simulation model synchronizes the events in a system being simulated. This is carried out by two methods.

Fixed Time Incrementing Method (FTIM)

Begin System in Initial State at a Fixed Point of Time

↓

Advance System Clock by a small fixed amount of time

↓

Update System from Events Occurred in Elapsed Time and Determine Resulting System State

↓

Repeat Last Two Steps As Many Times As Desired

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Model Synchronization

Next Event or Variable Time Incrementing Method (VTIM)

Begin System in Initial State at a Fixed Point of Time

↓

Advance System Clock by Variable Amount of Time Till Next Event Occurs

↓

Update System from Events Occurred in Elapsed Time and Determine Resulting System State

↓

Repeat Last Two Steps As Many Times As Desired

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Random numbers are needed to predict upcoming **events** which are then ordered in time.

Next jump from **one event to another** while **recording system changes** every time an event is reached.

Three Model Levels

Conceptual Model

- Definition of state variables, static/dynamic nature, entities, attributes, and events.

Specification Model

- Involve specifications
- Expressed in terms of inputs, outputs, simulation scheme, pseudocodes and equations

Computational Model

- Involve computer programs

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Verification and Validation

Verification

- Is the model built right?
- Does it serve the intended purpose?
- Computational model should be consistent with specification model

Validation

- Is the right model being built?
- Computational model should be consistent with the real system under consideration – does the model adequately describe the reality being modeled?
- Simulation output and system output should match – the decision makers must trust the model built.

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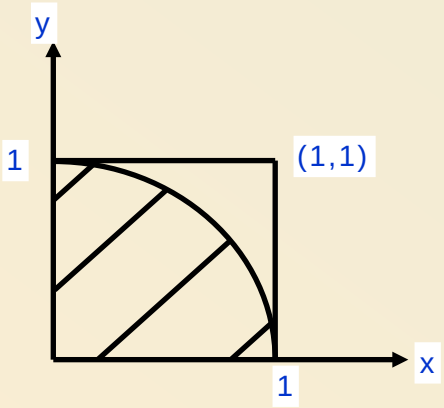
Monte Carlo Simulation

Estimate the value of π.



- Take two random numbers r1 and r2 uniformly distributed between 0 and 1.
- Let these two numbers represent x and y co-ordinates i.e. $x = r_1$ and $y = r_2$
- Check if point (x, y) is inside quarter-circle. Check $x^2 + y^2 > 1$ or $r_1^2 + r_2^2 > 1$
- Point (x, y) will always be inside unit square

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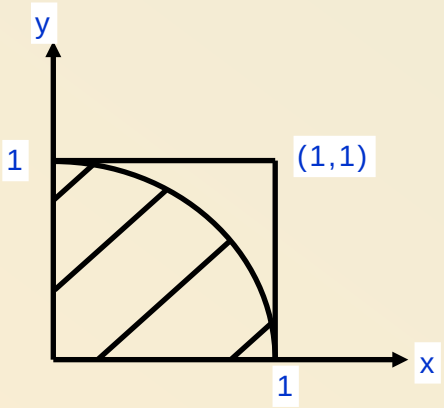
if $r_1^2 + r_2^2 > 1$ then point (x,y) is outside quarter circle

else point (x,y) is inside quarter circle

Monte Carlo Simulation

Estimate the value of π.



- **if $r_1^2 + r_2^2 > 1$ then point (x,y) is outside the quarter circle**
- **else point (x,y) is inside the quarter**
- **Point (x, y) will always be inside unit square**

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If we simulate for 10000 iterations, say p number of times the point falls inside the quarter circle.

Then, Estimated area of quarter-circle = (p/10000)*1 = p/10000

<u>Estimation of the value of π</u>

Area of a quarter-circle with unit radius = $\pi(1)^2/4 = \pi/4$

Equating this with the previous result, we get, $\pi/4 = p/10000$

So, the estimate of π = 4p/10000

Monte Carlo Simulation

- ***A problem on Prisoners' Dilemma***

- ***Two prisoners A and B are charged with a joint crime and are held incommunicado. If A confesses and B does not, A is given a reduced sentence of two years for cooperating with the authorities and B gets a 10 year prison term, and vice versa. If both confesses, each gets six years. If neither confesses, each will receive only a one-year sentence. Neither can know or even control the behavior of the other.***

- ***Design a simulation scheme in order to obtain the expected penalties of A and B under varying probabilities of prisoners taking different actions***

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Monte Carlo Simulation

Let us assume the following probabilities:

- **A confesses : a**
- **A does not confess : 1 – a**
- **B confesses : b**
- **B does not confess : 1 – b**

Choose 2 random numbers r_1 and r_2 between 0 and 1.

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If r_1 <= a;
Then *A confesses*
Else
A does not confess

If r_2 <= b;
Then *B confesses*
Else
B does not confess

Monte Carlo Simulation

- Depending on the outcomes, one of the following events will come about and corresponding penalties would result:

Event	Penalty
A confesses but B does not confess	A: 2 years and B: 10 years
A does not confess but B confesses	A: 10 years and B: 2 years
A confesses and B also confess	A: 6 years and B: 6 years
A does not confess and B does not confess	A: 1 year and B: 1 year

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- Finally, the expected penalty values for A and B are the average values of the corresponding penalties obtained in different iterations.

Monte Carlo Simulation

- **As a check, these expected penalty values can be compared with the following theoretically obtained values:**

Prisoner	Expected Penalty
A	a*(1 – b)*2 + (1 – a)*b*10 + a*b*6 + (1 – a)*(1 – b)*1 years
B	a*(1 – b)*10 + (1 – a)*b*2 + a*b*6 + (1 – a)*(1 – b)*1 years

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Discrete Event Simulation Example

An example from Queuing: In a bank with a single counter, Arrival is Poisson at 10 per hour and the Service time is exponential at 5 minutes per service.

Hence, Arrival Rate λ = 10 per hour

Service Rate μ = 60/5 = 12 per hour

How to simulate this queuing system?

- **Generate inter-arrival and service times from exponential distribution – how?**
- **Make use of Fixed or Variable Time Increment Method to obtain and record events.**
- **From the simulation data generated, obtain steady state parameters such a L, Lq, W, and Wq.**

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Discrete Event Simulation Example

Queuing System

- Entities: Customers, Servers, Bank
- Attributes: Arrival rate, Service rate, Queue space etc.
- Activities: Arrival, Service
- Events: An arrival, A Service Completed.
- State Variables: Number of Customers in the System, Number of Customers in Queue

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Pseudorandom Numbers

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Generating Pseudorandom numbers

What are pseudo-random numbers?

- **Simulation experiments require a source of random-ness. Pseudo-random numbers serve this purpose.**
- **These numbers are samples from uniformly distributed random variables between some specified interval.**
- **They have equal probability of occurrence as in the toss-up of an unbiased coin.**
- **A random number generating scheme attempts to produce a sequence of numbers between 0 and 1 which simulates the ideal properties of uniform distribution and independence as closely as possible.**

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Generating Pseudorandom numbers

- What is "pseudo" about pseudorandom numbers?
- Pseudorandom numbers are generated from a fast and deterministic method of generating a sequence of numbers that have the property of being random.
- Pseudo is used because the numbers are obtained from the use of a random number generator.
- Although pseudo means false, false random numbers are not generated by such a number generator.
- Here "pseudo" implies that generation of random numbers by a known method removes the potential for true randomness. If the method is known, the set of random numbers can then be replicated.

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Linear Congruential method of pseudorandom number generation

Linear Congruential method produces a sequence of integers X_i between 0 and (m–1) according to following recursive relationship:

$$X_{i+1} = (aX_i + c) \bmod m, \qquad \text{where } i = 0, 1, 2, \ldots$$

- **The initial value X_0 is called a seed,**
- **a is called the constant multipler,**
- **c is the increment, and**
- **m is the modulus.**

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Linear Congruential method of pseudorandom number generation

$X_{i+1} = (aX_i + c) \bmod m$, where $i = 0, 1, 2, \ldots$

If c is not equal to zero, then the method is called a **mixed congruential method**.

When c=0, the form is known as the **multiplicative congruential method**.

The selection of the values a, c, m, and X_0 drastically affects the **statistical properties** and the **cycle length**.

Variations of the equation are quite common.

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Linear Congruential method of pseudorandom number generation

- An example $X_0 = 27$, $a = 17$, $c = 43$, $m = 100$
- Integer values generated will all be between 0 and 99. Random integers are generated rather than random numbers.
- The numbers will be uniformly distributed between 0 and 99. To turn them to random numbers we have to divide each of them by m.

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- $X_0 = 27$
- $X_1 = (17*27 + 43) \bmod 100 = 502 \bmod 100 = 2$
- $X_2 = (17*2 + 43) \bmod 100 = 77 \bmod 100 = 77$
- $X_3 = (17*77 + 43) \bmod 100 = 1352 \bmod 100 = 52$
- Hence, $R_1 = 2/100 = 0.02$ $R_2 = 77/100 = 0.77$ $R_3 = 52/100 = 0.52$

Multiplicative Congruential method of pseudorandom number generation

- Multiplicative congruential generator consists of computing the following recurrent relation:

$$X_{i+1} = (a*X_i) \bmod m, \qquad \text{where } i = 0, 1, 2,\ldots$$

- Where
- X_i is the ith pseudo-random number,
- a is a constant multiplier, and,
- mod m denotes computation of the remainder (less than m) after repeated division of $a*X_i$ by m.
- This remainder is then set equal to the next pseudo-random number X_{i+1}.
- The process starts with an initial seed value of X_0

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Multiplicative Congruential method of pseudorandom number generation

Choice of m, X_0, and a

- The resulting number coming out of the modulus arithmetic, can only be one of 0, 1, 2, 3, …., m-1. Thus eventually the series would repeat itself. To prevent this, m should be one more than the largest integer that can be held in one word of computer being used. For word length of 32 bits, m is $(2^{31} – 1) + 1$.
- The seed X_0 must be relatively prime to m. As m is a power of 2, any odd positive integer for X_0 would do.
- The constant multiplier a should also be relatively prime to m – i.e. it should also be an odd integer. A good choice of a is $8*k \pm 3$.

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Testing Randomness of Pseudorandom numbers

- A random variate is obtained by transformation on a uniform variate.
- It is therefore necessary to test statistically whether the initial set of numbers are generated from a uniformly distributed set of number in a (0,1) interval.
- When generating random numbers, certain problems or errors can occur.
- These errors are usually departures from ideal randomness.

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Testing Randomness of Pseudorandom numbers

Errors, or departure from ideal randomness, are all related to:

- **The generated number may not be uniformly distributed.**
- **The generated number may be discrete-valued instead of continuous valued.**
- **The mean of the generated numbers too high or too low.**
- **The variance of the generated numbers too high or too low.**
- **There may be cyclic variations such as autocorrelation between numbers, numbers successively higher or lower than the adjacent numbers, or several numbers above the mean followed by several numbers below the mean.**

- **These departures from ideal randomness must be avoided in a good pseudorandom number generator. The tests for randomness are devised to check for these.**

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Testing Randomness of Pseudorandom numbers

Tests for checking randomness

- **The frequency test: The uniformity of digits generated are tested with the help of a goodness of fit procedure such as the chi-square test or the Kolmogorov-Smirnov test.**
- **The gap test: This test statistically analyze the number of digits that occur between the generation of a specified digit.**
- **The runs test: This test checks for sequence of digits that successively falls above or below 0.50 or checks to see if unusual runs up or down occur in successive numbers.**
- **The poker test: It examines individual digits to see if a single number or combination of numbers form pairs, threes etc. in successive sets of five random numbers.**

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Testing Randomness of Pseudorandom numbers

- **Serial test: This test checks the randomness of successive numbers in a sequence.**
- **Product Test: This test checks for independence or correlation between sequences of random numbers.**
- **Autocorrelation Test: This test examines the autocorrelation patterns or relationships between numbers in a sequence of random numbers - i.e. whether a number is dependent on another number in the sequence.**
- **Maximum test: The maximum test examines random number sequences to find out abnormally low or high numbers.**

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The Frequency Test

- Frequency or uniformity test counts how often numbers in a given range occur in the sequence to ensure that the numbers are uniformly distributed.
- There should be no favorite number and no number should occur more frequently that from expected chance variations.
- In 1000 two-digit numbers, we should expect about 100 numbers from 00 to 09, 100 numbers from 10 to 19 etc. – non-randomness in data may be suspected if it is say 30 or 200!
- The uniformity of digits generated are tested with the help of a goodness of fit procedure such as the chi-square test or the Kolmogorov-Smirnov test.
- The pseudo-random numbers must pass the frequency test among others.

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Simulation Example

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Simulation Example

- **Packages arrive at a Truck terminal with inter-arrival times according to the following probability values:**

Inter-arrival Time	3 minutes	4 minutes	5 minutes
Probability	0.3	0.35	0.35

- **A Truck arrives with inter-arrival times given as per the following probability values:**
-

Inter-arrival Time	10 minutes	15 minutes	20 minutes
Probability	0.3	0.4	0.3

- **Each Truck capacity is 10 and the number of packages already in the Truck is equally likely to be any number from 0 to 10.**
- **Additional packages beyond 10 are not allowed – these packages need to wait for the next Truck.**
- **Random Nos for Package Arrival: 38, 55, 17, 46, 22, 43, 69, 72, 24, 22, 61, 96, 29, 12, 03, 48**
- **Random Nos for Truck Arrival: 28, 88, 18, 51, 29**
- **Random Nos for Packages already in Truck: 04, 92, 85, 44, 45**

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Random Number Allocation

- **2-digit random number allocation to Package arrival, Truck arrival, and No. of Packages in a Truck.**

Package Arrival

Package Inter-arrival Time	Probability	Cumulative Probability	Random Numbers Allocated
3 minutes	0.30	0.30	00-29
4 minutes	0.35	0.65	30-64
5 minutes	0.35	1.00	65-99

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Truck Arrival

Truck Inter-arrival Time	Probability	Cumulative Probability	Random Numbers Allocated
10 minutes	0.30	0.30	00-29
15 minutes	0.40	0.70	30-69
20 minutes	0.30	1.00	70-99

Random Number Allocation

Packages in a Truck

Packages in a Truck	Probability	Cumulative Probability	Random Numbers Allocated
01	0.10	0.10	00-09
02	0.10	0.20	10-19
03	0.10	0.30	20-29
..	……	……	……..
..	……	……	……..
09	0.10	0.90	80-89
10	0.10	1.00	90-99

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Simulation for One Hour

- **Simulation for one hour is to be carried out to find out how many packages are still waiting for a truck after one hour.**

- **Generation of Arriving Packages:**

Package Inter-arrival Time	Probability	Cumulative Probability	Random Numbers Allocated
3 minutes	0.30	0.30	00-29
4 minutes	0.35	0.65	30-64
5 minutes	0.35	1.00	65-99

Random Number:	38	55	17	46	22	43	69	72	24	22	61	96	29	12	03	48
Inter-arrival Time	4	4	3	4	3	4	5	5	3	3	4	5	3	3	3	4
Cumulative Arrival Time	4	8	11	15	18	22	27	32	35	38	42	47	50	53	56	60

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Simulation for One Hour

- **Simulation for one hour is to be carried out to find out how many packages are still waiting for a truck after one hour.**

- **Generation of Truck Arrivals:**

Truck Inter-arrival Time	Probability	Cumulative Probability	Random Numbers Allocated
10 minutes	0.30	0.30	00-29
15 minutes	0.40	0.70	30-69
20 minutes	0.30	1.00	70-99

Random Number:	28	88	18	51
Truck Inter-arrival Time	10	20	10	15
Cumulative Arrival Time	10	30	40	55

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Simulation for One Hour

- **Simulation for one hour is to be carried out to find out how many packages are still waiting for a truck after one hour.**
- **Generation of Number of Packages already in Truck:**

Packages in a Truck	Probability	Cumulative Probability	Random Numbers Allocated
01	0.10	0.10	00-09
02	0.10	0.20	10-19
03	0.10	0.30	20-29
..	……	……	……..
..	……	……	……..
09	0.10	0.90	80-89
10	0.10	1.00	90-99

Random Number:	04	92	85	44
Number of Packages in Truck	01	10	09	05

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Simulation for One Hour

- Using the data, simulation for one hour is carried out to obtain the following results:

SL. No.	Time in Minutes	No. of Arriving Packages	No. of Waiting Packages	Packages already in a Truck	Packages could be allowed	Packages actually allowed	Packages still waiting
1	10*	2	2	01	9	2	0
2	30*	5	5	10	0	0	5
3	40*	3	8	09	1	1	7
4	55*	4	11	05	5	5	6
5	60	2	8	-	-	-	8

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* Times of Bus arrival

- It can be seen that 8 packages are still waiting for trucks at the end of simulation period of one hour.

Waiting Time Computation

- **Details of Arriving Packages within one hour:**

Random Number:	38	55	17	46	22	43	69	72	24	22	61	96	29	12	03	48
Inter-arrival Time	4	4	3	4	3	4	5	5	3	3	4	5	3	3	3	4
Cumulative Arrival Time	4	8	11	15	18	22	27	32	35	38	42	47	50	53	56	60

- **Details of Package Loading into Trucks:**

Truck Arriving at	10	30	40	55
Packages Actually Loaded	2	0	1	5

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- **Package Waiting Time Computation**

Cumulative Arrival Time	4	8	11	15	18	22	27	32	35	38	42	47	50	53	56	60
Truck Load Time	10	10	40	55	55	55	55	55	-	-	-	-	-	-	-	-
Waiting Time	6	2	29	40	37	33	28	23	-	-	-	-	-	-	-	-

Generation of Random Variates

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Random Variates

- A random variate is an artificially generated random variable.
- Input variables for simulation follow some specified probability distribution. To generate some samples from a given distribution, random numbers are drawn.
- Let F(X) be the cumulative density function of the input variable and R a random number. Then we have:

$$R = F(X)$$

Hence, $X = F^{-1}(R)$

Where X is the random variate.

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Exponential Random Variate

- The density function of the exponential distribution is given by: $f(t) = \lambda e^{-\lambda t}$ where λ is a positive constant characteristic of the system at hand
- For exponential interarrival time of customers, λ is the average number of arrivals per unit time.
- Hence, the cumulative probability distribution function: $F(t) = 1 - e^{-\lambda t}$.
- The average or the expected value is given by = $1/\lambda$
- To obtain a number of exponentially distributed random numbers $t_1, t_2, \ldots, t_n$, The cdf F can be equated with u, a generated uniform random number.
- So, $u = F(t)$ and Exponential Random Variate, $t = F^{-1}(u)$

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Exponential Random Variate

- Thus, $u = F(t) = 1 - e^{-\lambda t}$
- So that, $-\lambda t = \ln(1 - u)$
- Which gives the corresponding sample t as
- $t = -\ln(1 - u)/\lambda$
- Since, (1 – u) is as good a uniformly distributed random number between 0 and 1 as u is, we can replace (1 – u) by u itself.
- Thus, Exponential Random variate, $t_k = -\ln u_k/\lambda$

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Generating Random Variate

Inverse Function Method

- Consider a random variable x with the following probability density function: $f(x) = 2x^2$ Find its random variate.
- As $f(x) = 2x^2$, The cumulative distribution function is given by: $F(x) = 2x^3/3$
- Therefore, for the random variable x, $R = F(x) = 2x^3/3$
- The random variate x is thus generated by:
- $x = \sqrt[3]{(3R/2)}$
- Where R is a random number between 0 and 1.

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Generating Random Variate

Consider a random variable X with the following probability density function:

- **$f(x) = x$ for $0 <= x <= 1$;**
- **$f(x) = 2 – x$ for $1 < x <= 2$**
- **$f(x) = 0$ otherwise**
- **Find its random variate.**

- **The distribution is known as the triangular distribution with endpoints (0,2) and mode at 1. Its cdf is given by:**

- **$F(X) = 0$ for $x <= 0$;**
- **$F(x) = x^2/2$ for $0 < x <= 1$;**
- **$F(x) = 1 – ((2 – x)^2/2)$ for $1 < x <= 2$;**
- **$F(x) = 1$ for $x > 2$.**

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Random Variate Generation

- Therefore, for the random variable X, $0 <= X <= 1$,
- $R = X^2/2$
- And, for random variable X, $1 <= X <= 2$,
- $R = 1 - ((2 - X)^2/2)$
- This gives us a pointer to the random variate. X is thus generated by:

$X = \sqrt{(2R)}$ for $0 <= R <= 1/2$, and

$X = 2 - \sqrt{(2(1 - R))}$ for $1/2 < R <= 1$

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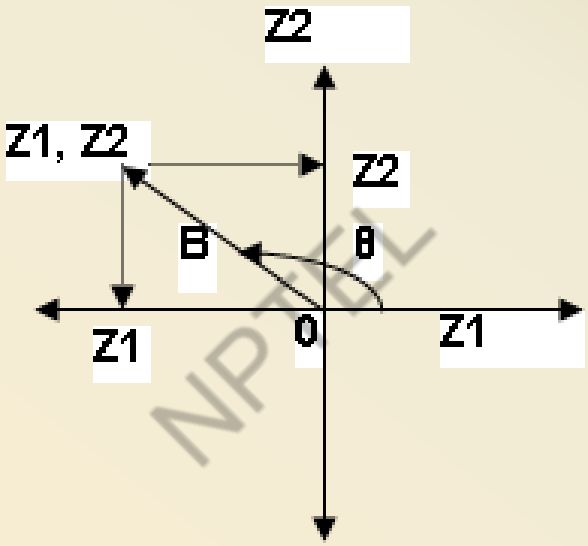
Normal Random Variate

- Consider two standard normal random variables Z1 and Z2 plotted as a point in the plane as shown below:



- Represented in polar co-ordinates, we have,
- $Z1 = B\cos\theta$ and $Z2 = B\sin\theta$

Normal Random Variate

- $B^2 = Z1^2 + Z2^2$ has chi-square distribution with 2 degrees of freedom - equivalent to an exponential distribution with mean 2.
- Thus the radius B can be generated as: $B = (-2\ln R)^{1/2}$
- By the symmetry of the normal distribution, we can assume that the angle is symmetrically distributed between 0 and 2π radians. Also radius B and angle θ are mutually independent.
- Using previous equations, two independent standard normal variates Z1 and Z2 can be generated directly from two independent random numbers R1 and R2.
- Hence, we have: $Z1 = (-2\ln R1)^{1/2}\cos(2\pi R2)$, and

  $Z2 = (-2\ln R1)^{1/2}\sin(2\pi R2)$

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Erlang Random Variate

An ERLANG random variable X with parameters (K, λ) can be shown to be the sum of K independent exponential random variables,

X_i (i = 1, 2, 3, …, K), each having a mean 1 / Kλ, that is

$$X = \sum_{i=0}^{K} X_i$$

Since, each X_i can be generated by the exponential random variates:

$$X_i = -\ln R_i / K\lambda,$$

An ERLANG random variate can be generated by:

$$X = \sum_{i=0}^{K} \frac{-1}{K\lambda} \ln R_i = \frac{-1}{K\lambda} \ln \prod_{i=1}^{K} R_i$$

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Rejection Method

- **The rejection method for obtaining sample from a given non-uniform distribution works by generating uniform random numbers repeatedly and accepting only those numbers that fulfill a given condition.**

- **This condition of acceptance need to be designed such that the accepted numbers appear to be drawn from the given distribution.**

- **For the rejection method to be applicable, the probability density function f(t) of the distribution must be nonzero only over a finite interval (P,Q*).***

- ***Let f(t) be bounded by an upper limit R.***

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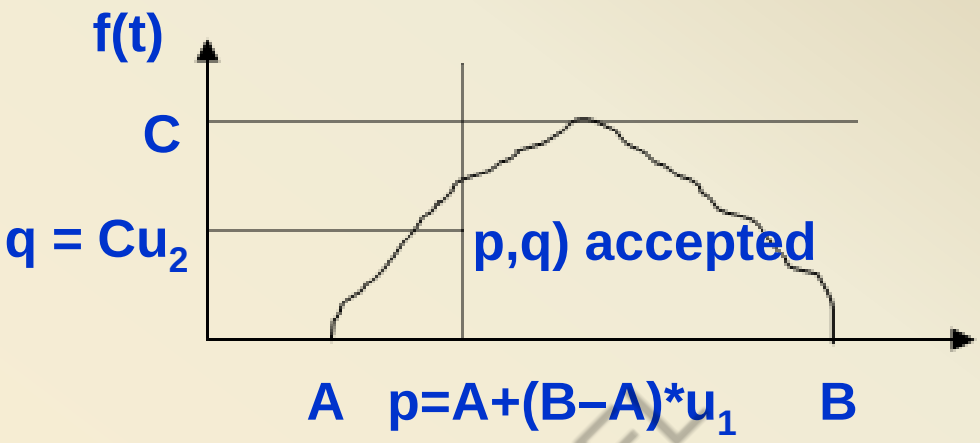
Rejection Method



The steps are as follows:

1) generate a pair of random numbers u_1, u_2 in the interval (0,1)

2) using u_1, locate point p on horizontal axis t as $p=A+(B-A)*u_1$

3) using u_2, locate point q on the vertical axis as $q = Cu_2$

4) if q > f(p) reject the pair and go to step 1); else accept p as the value of a sample from the desired distribution.

The method works only for a finite interval.

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THANK YOU

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